

1. Purpose

AnimaTruc provides a proprietary animation runtime for Minecraft Forge mods with:

- deterministic blending
- animation graph layers
- procedural modifiers
- block-ground contact IK
- automatic locomotion presets

2. Automatic Locomotion Presets

Supported automatic profiles:

- BIPED = 2 legs
- HEXAPOD = 6 legs
- OCTOPOD = 8 legs
- MYRIAPOD = more than 8 legs (centipede/millipede class)

Main entrypoint:

- io.hoyatla.animatruc.forge.preset.AutoGroundContactPresetFactory

Factory output:

- AutoGroundContactPreset
 - createIkModifier()
 - createGroundController()
 - detectionReport()

3. Anti False-Positive Leg Detection Strategy

Detection is score-based and rejects weak chains.

3.1 Naming score

Positive tokens:

- End: foot, paw, hoof, toe, claw
- Mid: lower_leg, shin, knee, tibia, calf
- Root: upper_leg, thigh, hip

3.2 Hierarchy constraints

A valid chain must resolve:

- end bone -> mid parent -> root parent

3.3 Exclusion filters

The chain is rejected when name tokens match:

- arm, hand, finger, wing, tail, head, antenna, horn, ear

3.4 Geometry sanity checks

The chain confidence is improved only if:

- upper and lower lengths are non-zero
- pivots are coherent for lower-body usage

3.5 Confidence threshold

Only chains above minimum confidence are accepted.
Default is 0.55 and configurable.

3.6 Safety caps

- maxDetectedChains limits total chain count
- maxMyriapodChains limits many-leg profiles

4. Recommended Bone Naming

Use clear names to maximize stable auto-detection:

- left_upper_leg, left_lower_leg, left_foot
- right_upper_leg, right_lower_leg, right_foot

Hexapod example:

- front_left_upper_leg / front_left_lower_leg / front_left_foot
 - mid_left_upper_leg / mid_left_lower_leg / mid_left_foot
 - rear_left_upper_leg / rear_left_lower_leg / rear_left_foot
- (and right side equivalent)

Myriapod recommendation:

- segment_00_left_upper_leg ... segment_00_left_foot
- segment_00_right_upper_leg ... segment_00_right_foot
- repeat by segment index

5. Runtime Integration Flow

Step 1: Load pack and read skeleton

Step 2: Detect preset via AutoGroundContactPresetFactory

Step 3: Add IK modifier to animator

Step 4: Every client tick, update ground controller and inject targets in AnimatorContext

Reference snippet:

```
Optional<AutoGroundContactPreset> preset =
AutoGroundContactPresetFactory.detect(pack.skeleton());
if (preset.isPresent()) {
    AutoGroundContactPreset p = preset.get();
    animator.addModifier(p.createIkModifier());
    GroundContactController controller = p.createGroundController();

    Map<String, Vec3f> targets = controller.update(entity, entity.getYRot(), partialTicks);
    AnimatorContext ctx = GroundContactContextAdapter.attachTargets(
        AnimatorContext.visibleNear(),
        targets,
        GroundContactContextAdapter.horizontalSpeed(entity)
    );
    animator.update(ctx, partialTicks);
}
```

6. Configuration Menu (Lightweight Machines)

File:

- config/animatruc-client.toml

In-game menu:

- AnimaTruc Presets

Toggles:

- enableAutoDetection
- enableBipedPreset
- enableHexapodPreset
- enableOctopodPreset
- enableMyriapodPreset
- enableGroundRaycast
- performanceMode = FULL / BALANCED / LIGHT
- minDetectionConfidence

Performance recommendation:

- FULL: high quality, higher CPU
- BALANCED: default
- LIGHT: low-end hardware mode

7. Client/Server Side Rules

- Ground-contact raycasts and IK target generation should run client-side for visual animation.
- Keep gameplay authority on server and do not attach combat/game logic to visual IK output.

8. Myriapod Specific Notes

- Keep chain cap enabled (maxMyriapodChains).
- Use shorter step distance and lift height.
- Use LIGHT mode on low-end hardware.
- Keep naming regular by segment index and side.

9. Debug Checklist

If legs are misdetected:

- verify clear leg naming
- ensure 3-bone chain hierarchy exists
- lower confidence threshold only when naming cannot be changed
- inspect detectionReport warnings
- disable problematic profile in config menu

10. API Locations

Core detection:

- animatruc-proprietary-core/src/main/java/io/hoyatla/animatruc/core/preset/

Forge preset/config:

- animatruc-forge/src/main/java/io/hoyatla/animatruc/forge/preset/
- animatruc-forge/src/main/java/io/hoyatla/animatruc/forge/config/
- animatruc-forge/src/main/java/io/hoyatla/animatruc/forge/ground/

End of manual.